5 MARKS

1. Explain the basic characteristics of a Multi-Agent System with a suitable example.
2. Explain the role of agents in a Multi-Agent System.
3. Differentiate between a single-agent system and a Multi-Agent System
4. Analyze the importance of communication among agents in a Multi-Agent System
5. Design a simple Multi-Agent System for managing traffic at a road junction. Explain the role of each agent.
6. Explain cooperation and coordination among agents with a suitable example.
7. Explain the process of negotiation between two agents.
8. Differentiate between cooperation and coordination in Multi-Agent Systems.
9. Analyze how negotiation can help resolve conflicts between agents.
10. Propose a simple conflict-resolution strategy for two agents having different goals.
11. Explain the concept of Swarm Intelligence with a suitable example.
12. Explain the basic concept of Distributed Artificial Intelligence.
13. Differentiate between Swarm Intelligence and traditional centralized AI.
14. Analyze how simple agents can collectively solve a complex problem using Swarm Intelligence.
15. Design a simple swarm-based system for finding the shortest path between two locations.
16. Explain the use of Multi-Agent Systems in smart city applications
17. Explain how Multi-Agent Systems can be used in robotics.
18. Analyze the benefits of using multiple agents for traffic management in a smart city.
19. Explain how Multi-Agent Systems can improve healthcare services.
20. Design a simple Multi-Agent System for managing robots in a warehouse.
21. Explain the basic working of Reinforcement Learning with a suitable example.
22. Explain the main components of a Markov Decision Process (MDP).
23. Differentiate between Reinforcement Learning and supervised learning.
24. Analyze the role of state, action and reward in an MDP.
25. Formulate a simple MDP for a robot moving from one room to another. Identify its states, actions and rewards.
26. Explain the concept of a reward function in Reinforcement Learning.
27. Differentiate between immediate reward and long-term value.
28. Analyze how positive and negative rewards influence the behavior of an agent.
29. Design a simple reward function for an autonomous robot navigating through a grid environment.
30. Explain the basic steps of the Q-Learning algorithm.
31. Differentiate between Q-Learning and Deep Reinforcement Learning.
32. Analyze how Q-values help an agent select the best action.
33. Design a simple Q-Learning approach for an agent learning to reach a destination in a grid.

5 MARKS

34. Explain the basic steps of the Q-Learning algorithm.
35. Explain the role of a Q-table in Q-Learning.
36. Explain exploration and exploitation in Reinforcement Learning.
37. Explain the exploration-exploitation trade-off with a suitable example.
38. Differentiate between exploration and exploitation with respect to agent decision-making.
39. Analyze why an agent cannot always select the action with the highest known reward.
40. Propose a simple strategy to balance exploration and exploitation in a learning agent.
41. Explain the concept of Explainable AI (XAI) with a suitable example.
42. Explain why explainability is important in autonomous AI systems.
43. Analyze how Explainable AI can improve trust in autonomous systems.
44. Differentiate between an explainable AI system and a black-box AI system.
45. Design a simple explainable AI approach for an autonomous vehicle decision-making system.
46. Explain any three ethical issues related to autonomous AI systems.
47. Explain bias and fairness in Artificial Intelligence with suitable examples.
48. Analyze how biased training data can affect the decisions of an AI system
49. Explain the importance of accountability in autonomous AI systems.
50. Propose simple measures to reduce bias and improve fairness in an AI-based decision system.
51. Explain the concept of Edge AI and its advantages.
52. Explain the major privacy and security challenges in autonomous IoT systems.
53. Analyze the importance of privacy when AI systems process data from IoT devices.
54. Differentiate between cloud-based AI and Edge AI for IoT applications.
55. Design a simple secure Edge AI system for monitoring smart-home devices.
56. Explain the concept of Generative AI agents with a suitable example.
57. Explain how Generative AI can support autonomous agents in performing tasks.
58. Analyze the role of Generative AI in future autonomous systems.
59. Differentiate between a traditional AI agent and a Generative AI agent.
60. Propose a simple application of a Generative AI agent for a smart city and explain its benefits.